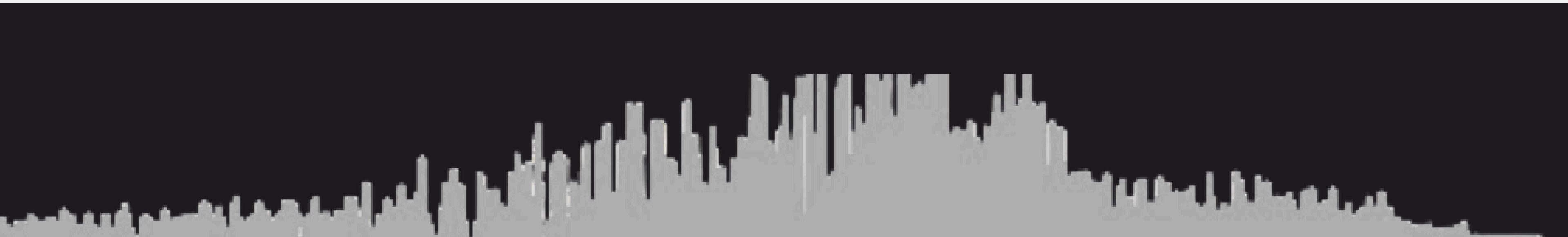


Frequency-Domain Patterns Across Earthquake Stations

Emiliano Hernández Castillo & Tarvin Kaur Arora



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Introduction

Research Question

- Do different earthquakes produce similar frequency-domain patterns at the same seismic stations?
- Can Self-Organizing Maps (SOMs) identify consistent spectral clusters across multiple earthquakes?

UNSUPERVISED LEARNING

SPECTRAL ANALYSIS

FOURIER TRANSFORM

MACHINE
LEARNING

Relevance

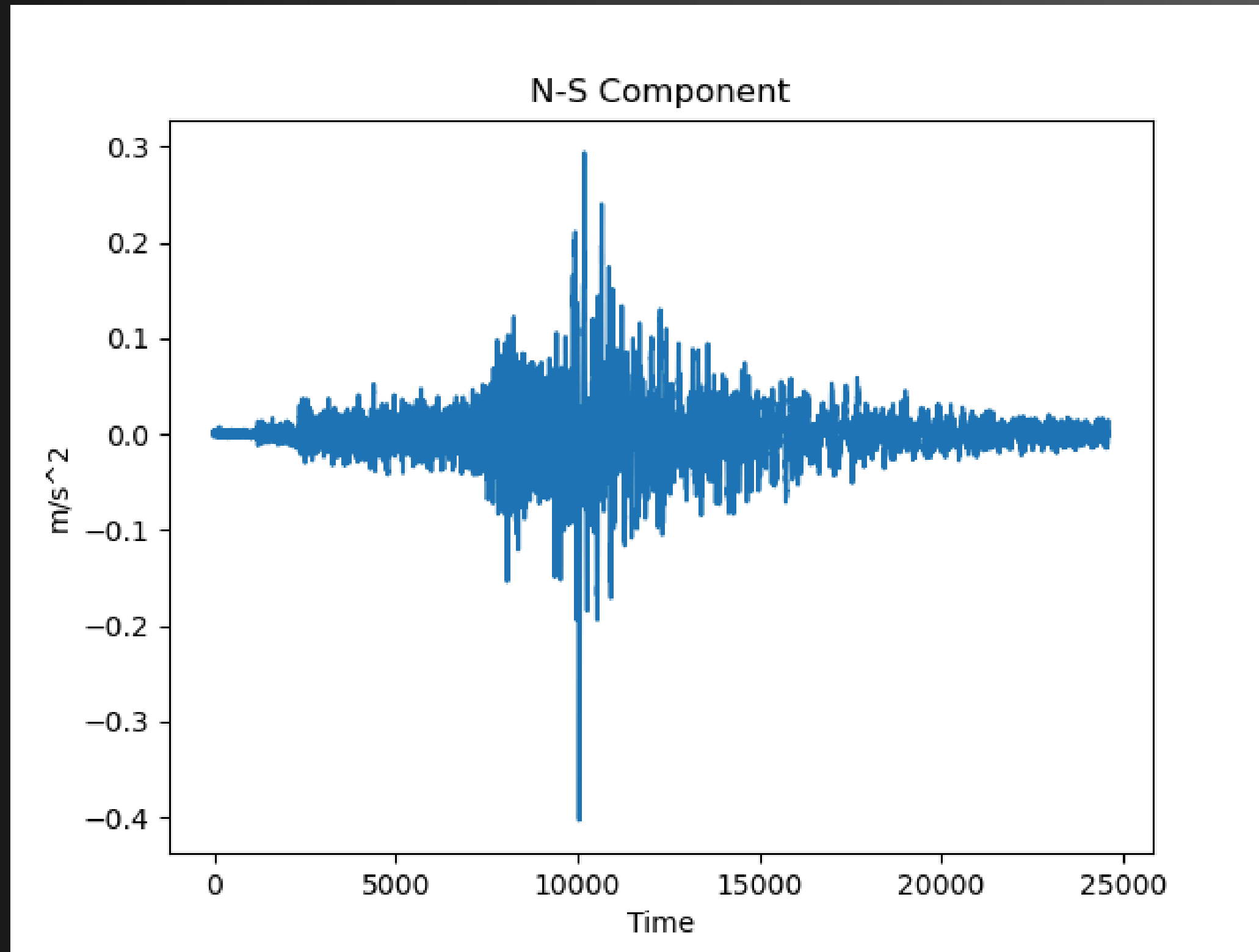
Why did we choose this project?

- Understanding earthquakes through spectra is essential for **seismic hazard prevention** in seismic zones, modelling, geophysics and engineering.
- We can demonstrate **how machine learning and signal processing reveal patterns** in geophysical data which sits at the core of our class.
- Can this spectral analysis help prevent disasters through **earlier warning signs, risk evaluation and improved resistance design?**
- Spectral information can help understand energy content per frequency which is a popular exploration in **earthquake prediction and earthquake resistant engineering.**
- Understanding spectral patterns creates reusable foundations that **enable more advanced models to be developed in the field.**

Dataset Description

- **Data Source:** Red Acelerográfica del Instituto de Ingeniería de la Universidad Nacional Autónoma de México (RAII UNAM).
- Data base contains > 21,000 accelerograms since 1964.
- We have chosen to focus on **3 earthquakes in 2022**.
- Each earthquake has accelerograms from **5 stations across Mexico** with approximately 26,000 data points through time ($dt = 0.005$) → 15 seismograms total.
- Each accelerogram contains **three components of ground acceleration (NS, EW, V)**, earthquake magnitude, focal depth, sampling frequency, lat/lon, recorder specifications, etc.
- For our project we focused on the three-component accelerations provided.

Dataset Example



(RAII UNAM, 2022)

1

**WINDOWING &
STANDARDIZATION**

3

**SOM
CLUSTERING**

5

**CROSS-EVENT
COMPARISON**

METHODOLOGY

2

**FAST FOURIER
TRANSFORM**

4

**PATTERN
VISUALIZATIONS**

Data Pre-Processing

- Data files downloaded from RAI UNAM were unreadable, thus all data had to be manually extracted into a .csv file.
- Each accelerogram was split into a non-overlapping 256-sample window, to isolate frequency content and to be able to properly inspect various spectrograms at once.
- The Fast Fourier Transform was computed for each window and their respective power spectra.
- Each spectrum was standardised so that the SOM focuses on its shape and not solely on its power.



(RAI UNAM, 2022)

Machine Learning

- Power spectra obtained from the Fast Fourier Transform reveal how the energy of an earthquake is distributed across different frequencies. Since we aim to find similarities across different earthquakes for a general classification, the energy content of a frequency is appropriate for our study.
- Self Organising Maps (SOMs) was chosen as a method as through the creation of hundreds of spectrograms, we considerably transitioned from a low dimensional data set to a high dimensional data set. Self Organising maps helped reduce our dimensions into a 2D map where similar spectrograms were grouped together which revealed data patterns across our accelerograms events. Subsequently, we used U-matrices to compare BMUs per station to locally compare.

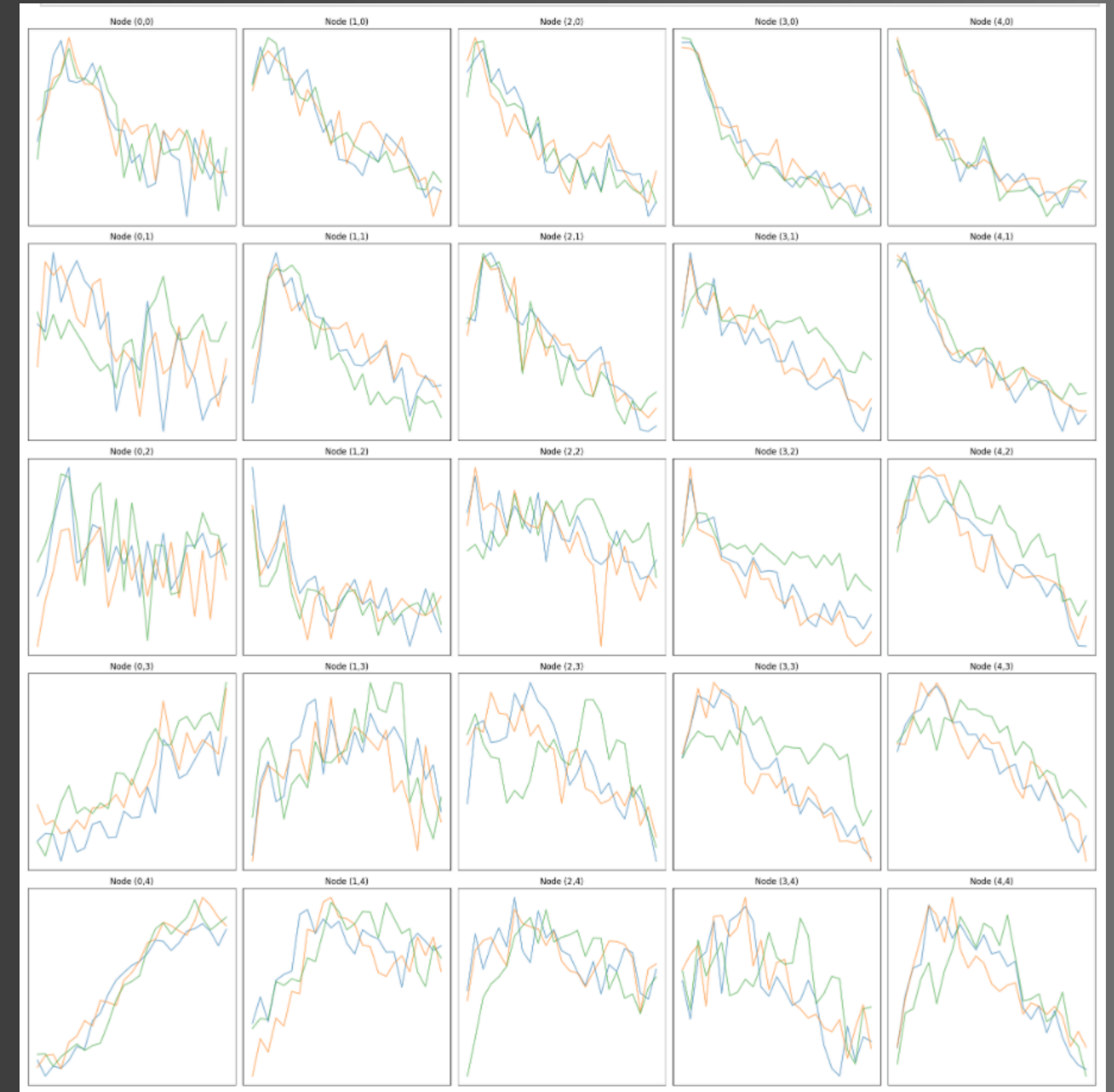


(RAII UNAM, 2022)

SOM Model training

Self Organising Maps

- All windowed and standardized spectra were stacked into a single data set which was then used as the input into a 5 x 5 Self Organising Map
- Iteratively, each BMU was placed closer to its cluster that best describes the nearby spectrograms
- All BMU across stations and earthquakes were mapped onto the SOM to find if there were any preliminary similarities in spectra patterns.



Data Analysis

- Principal Component Analysis (PCA) was also used to capture the dominant modes of spectra clusters created by the SOM. PCA was implemented to further reduce the dimensionality and capture the dominant modes (ie. largest variance) seen in each of the SOMs best matching unit (BMUs).
- Similarity matrices were taken to create a comparison between clusters provided. This method was useful to quantitatively and qualitatively observe broader spatial patterns observed between all 3 earthquakes.
- All steps taken will lead to the power spectras most commonly seen throughout all 3 studied earthquakes which set up a base for future predictions and our conclusion.

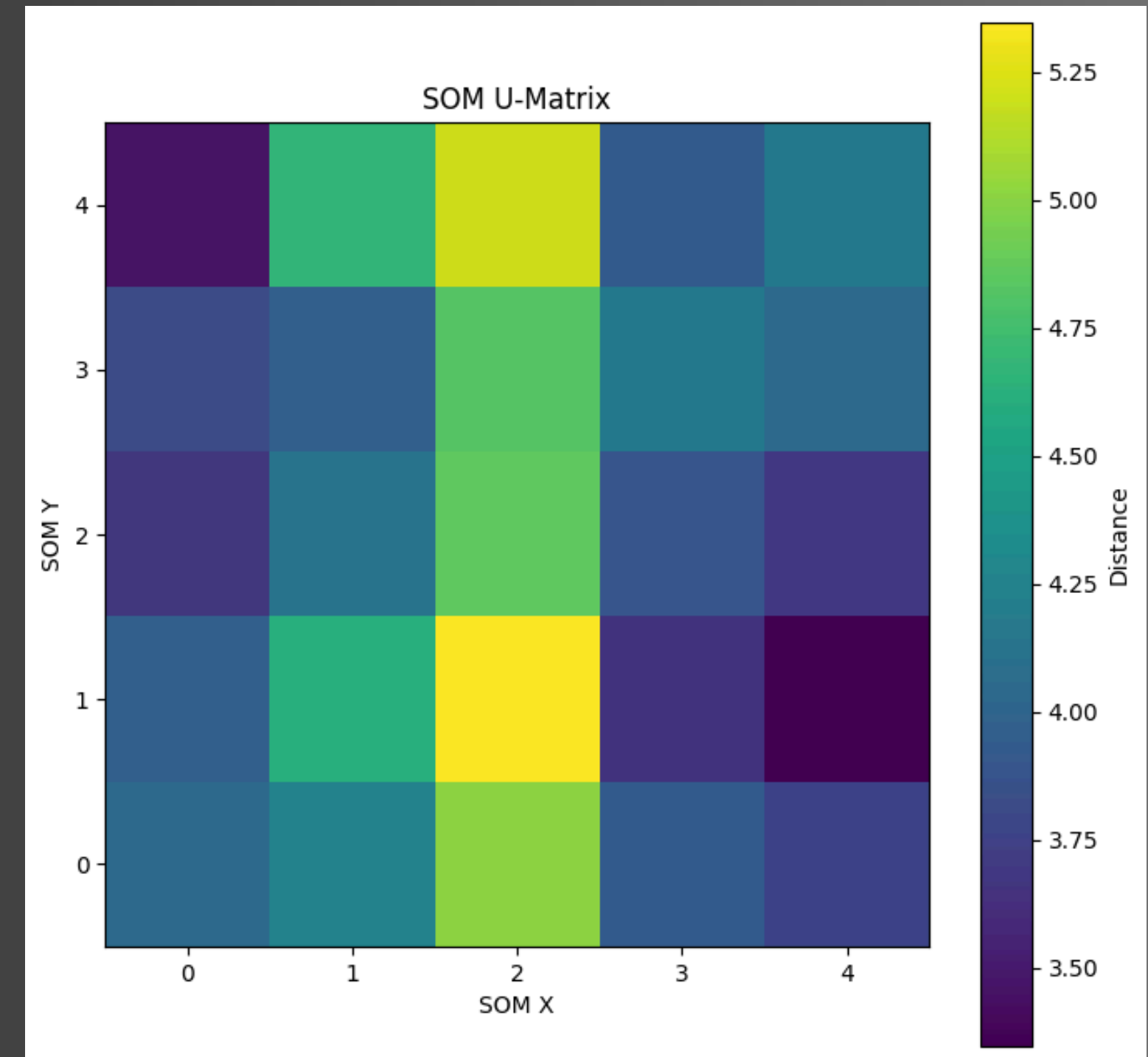


(RAII UNAM, 2022)

Data Analysis

U-Matrices

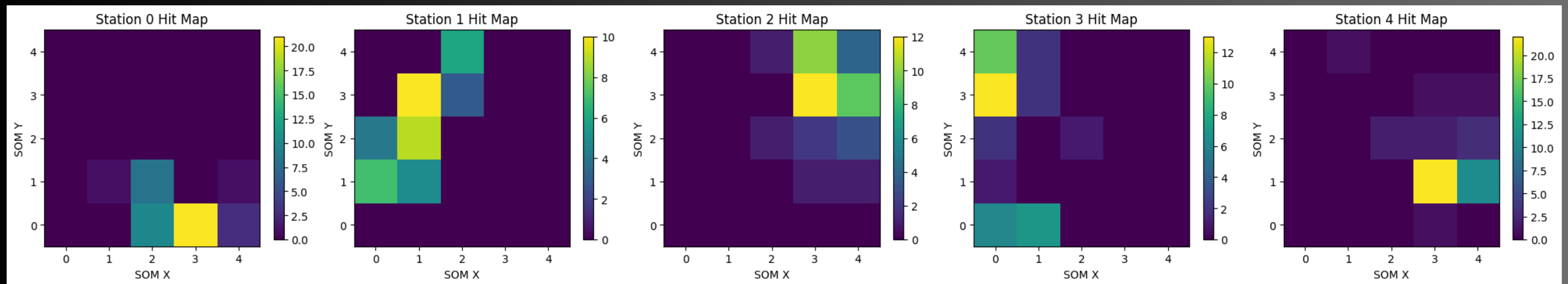
- U-Matrix was another visualization tool used to visualize the distances between neighbouring SOM nodes.
- Each SOM node represents a characteristic spectral pattern learned from the FFT windows.
- The U-Matrix shows how different SOM nodes relate to one another. Bright regions mark strong differences in spectral shape — essentially the boundaries between clusters — while darker regions indicate smooth, consistent spectral patterns within a cluster. The distinct bright and dark regions in our U-Matrix validate that the SOM found real, meaningful spectral regimes in the earthquake data.



Data Analysis

Station Hit Maps

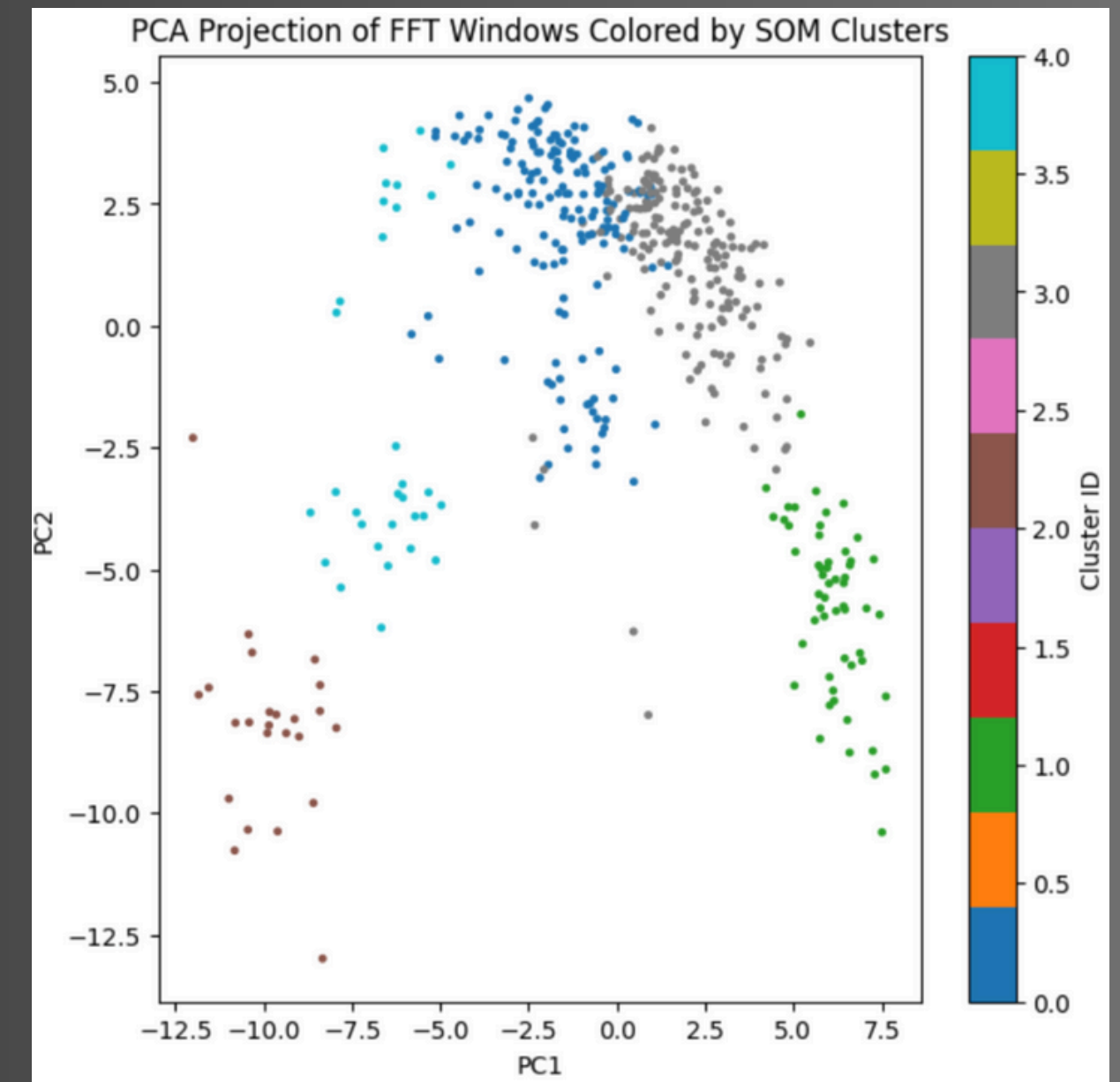
- These hit maps tell us which spectral regimes each station experienced.
- Stations with similar hit map patterns recorded similar shaking characteristics, which can reflect comparable geological conditions or similar distances from the earthquake.
- Stations with distinctly different hit map patterns show unique site responses, such as stronger low-frequency amplification or more high-frequency energy.



Data Analysis

Principal Component Analysis

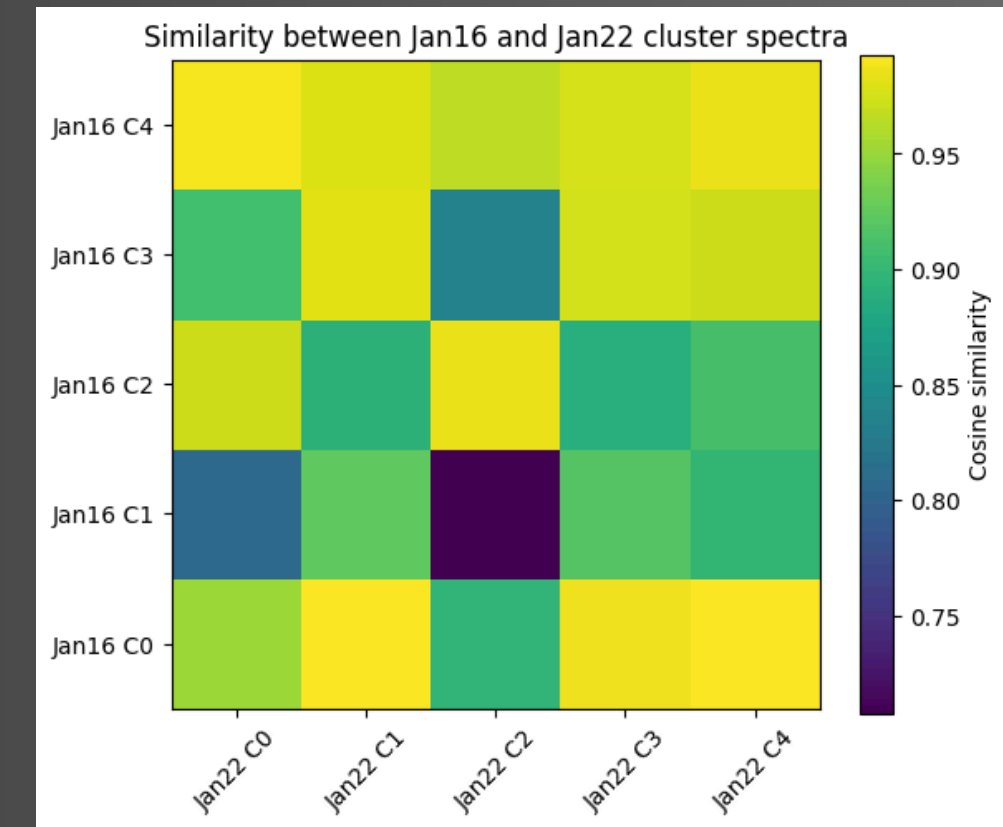
- Data obtained from SOMs was inputted into PCA to reduce dimensionality of data.
- The first 2 components of the data was kept as it represented the bulk of the data variance.



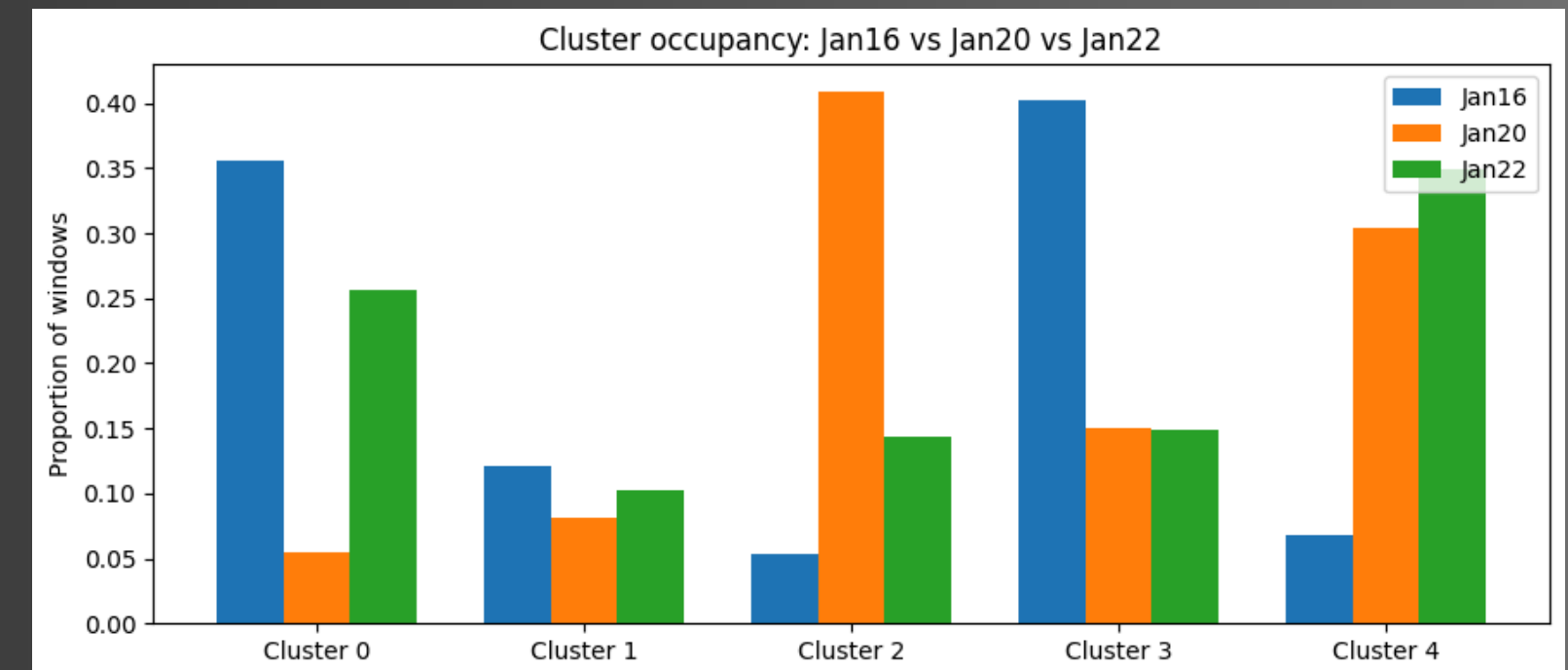
Comparison of Spectral Analysis

Similarity

- We measured the similarity between every pair of clusters using cosine similarity, which quantifies how close two high-dimensional spectral shapes are. The resulting matrix visualizes these pairwise similarities.

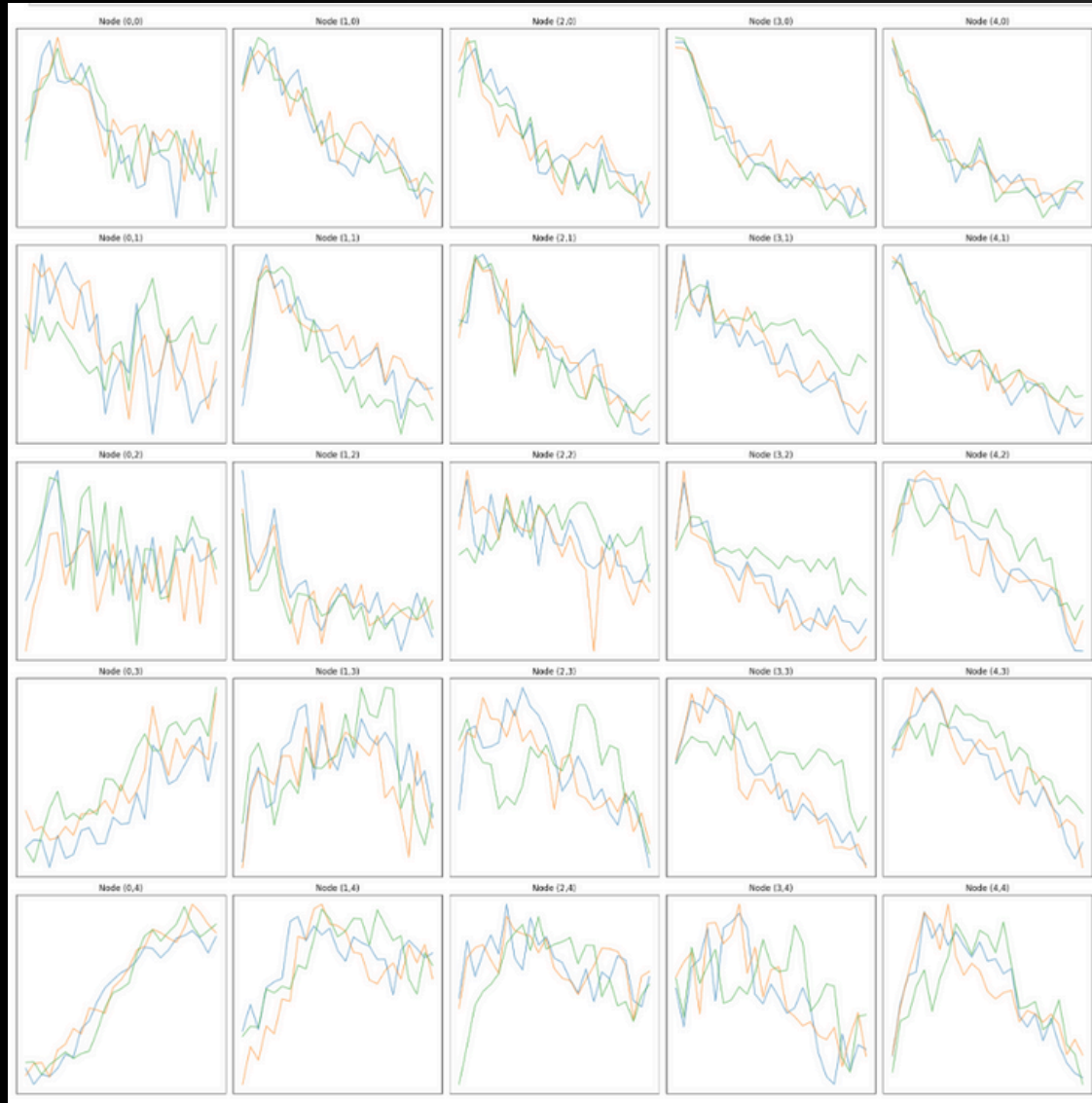


- For each earthquake, every FFT window was assigned to one of the SOM clusters. We then counted how many windows fell into each cluster and converted these counts into proportions. The bar chart visualizes the relative occupancy of spectral clusters across earthquakes.

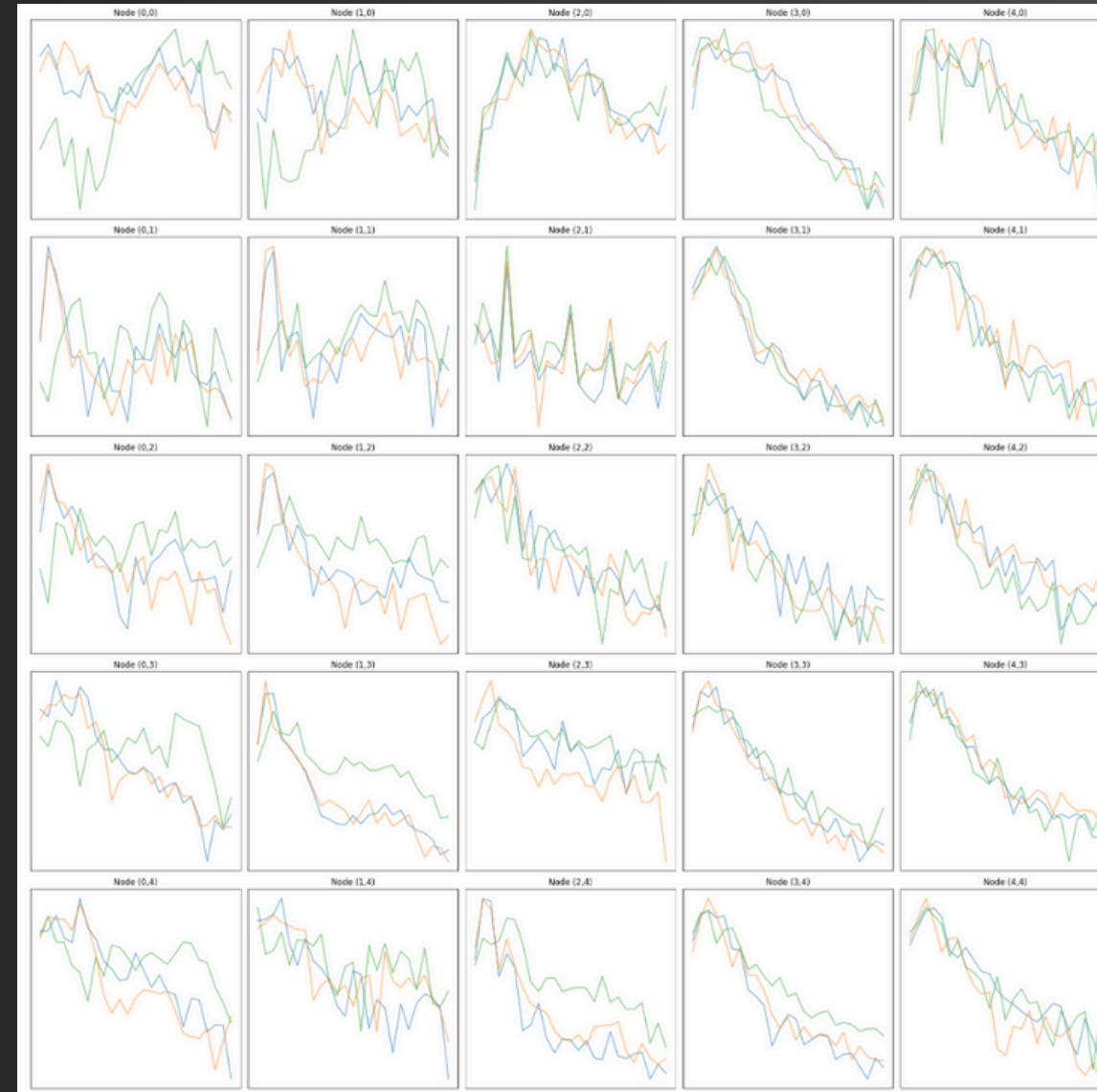


Results

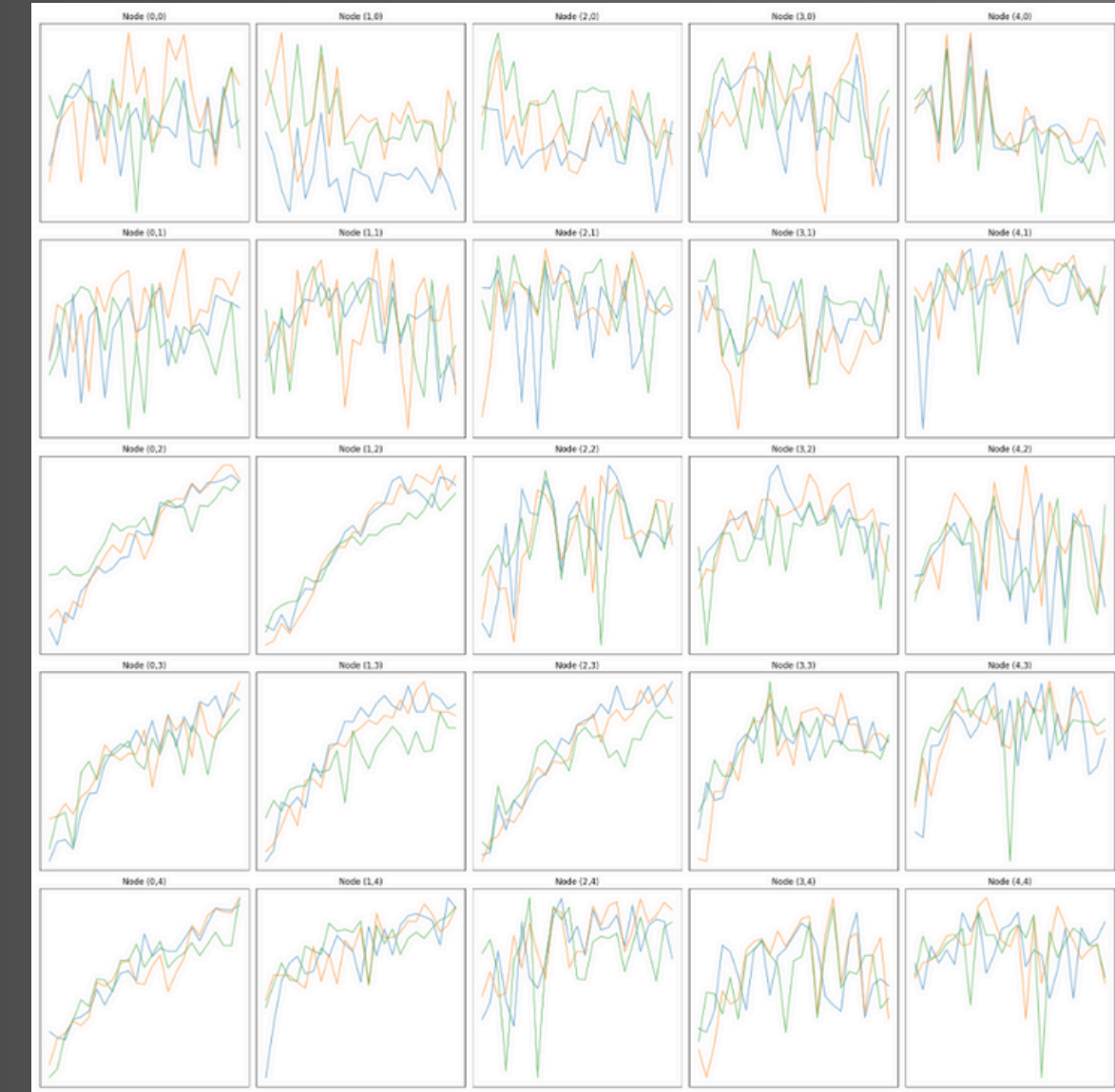
Resulting BMU's



January 16, 2022



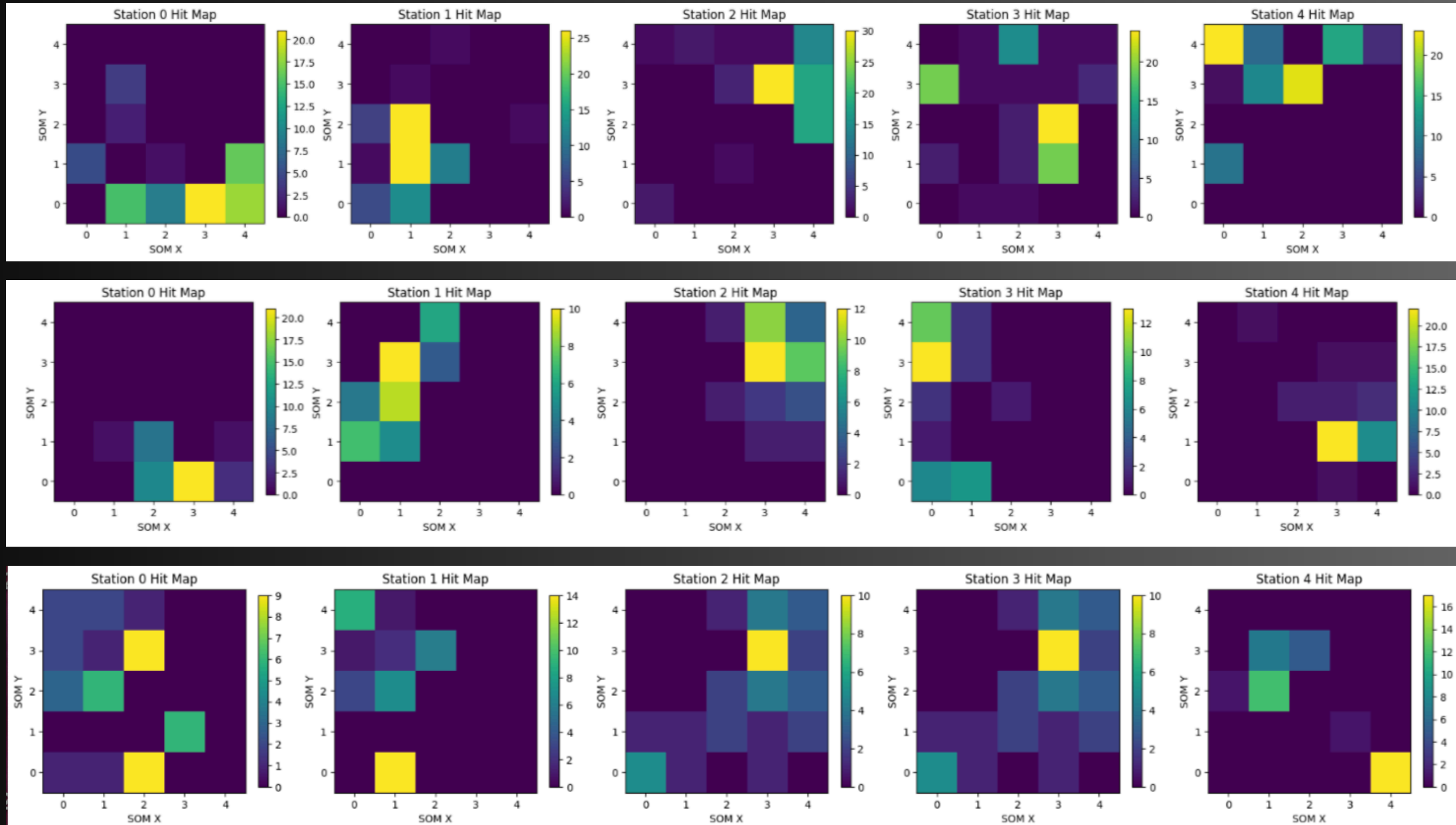
January 20, 2022



January 22, 2022

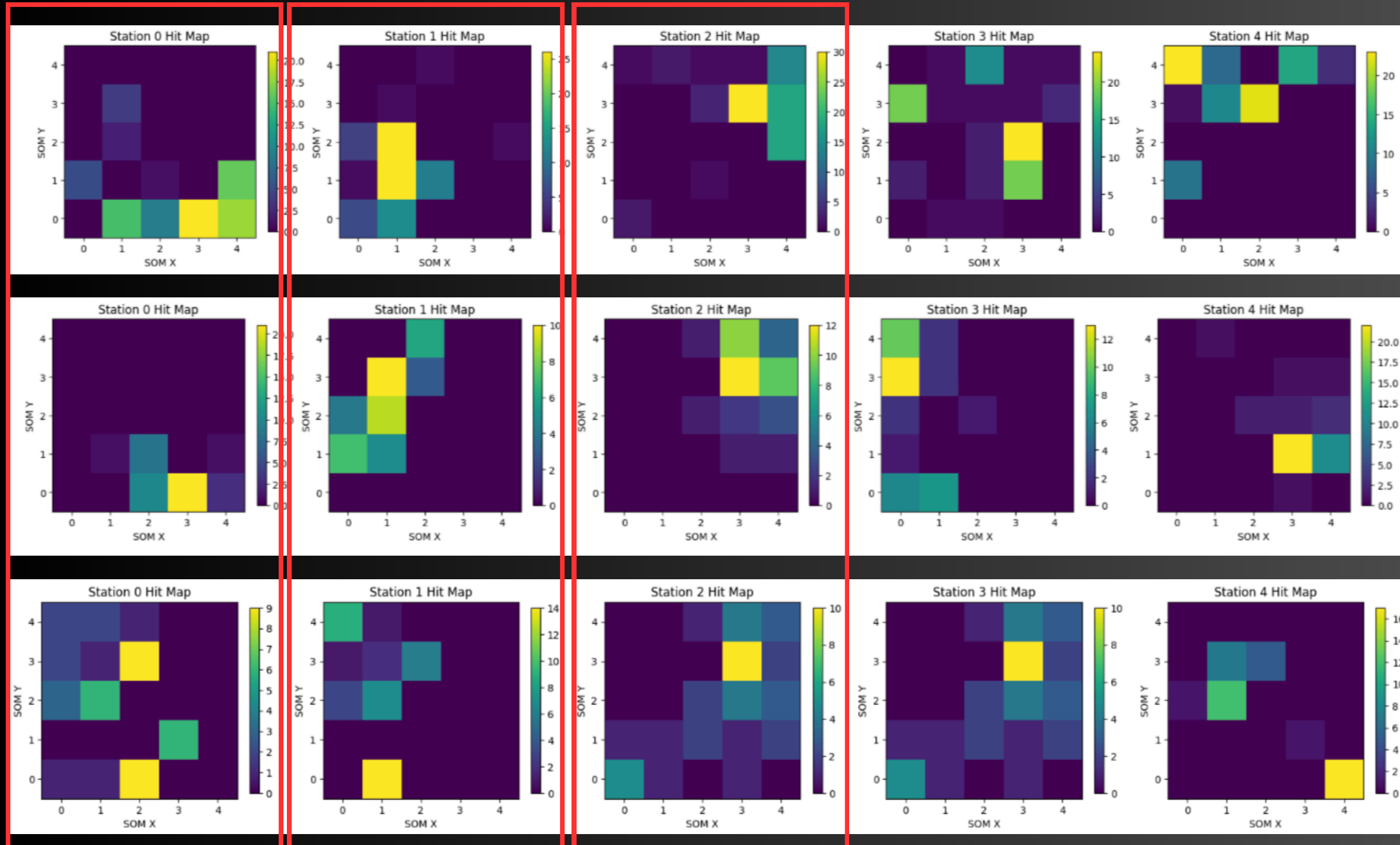
The SOM node spectra exhibit consistent frequency-domain shapes, indicating that the network reliably identifies recurring spectral regimes.

Resulting Station U-Matrices



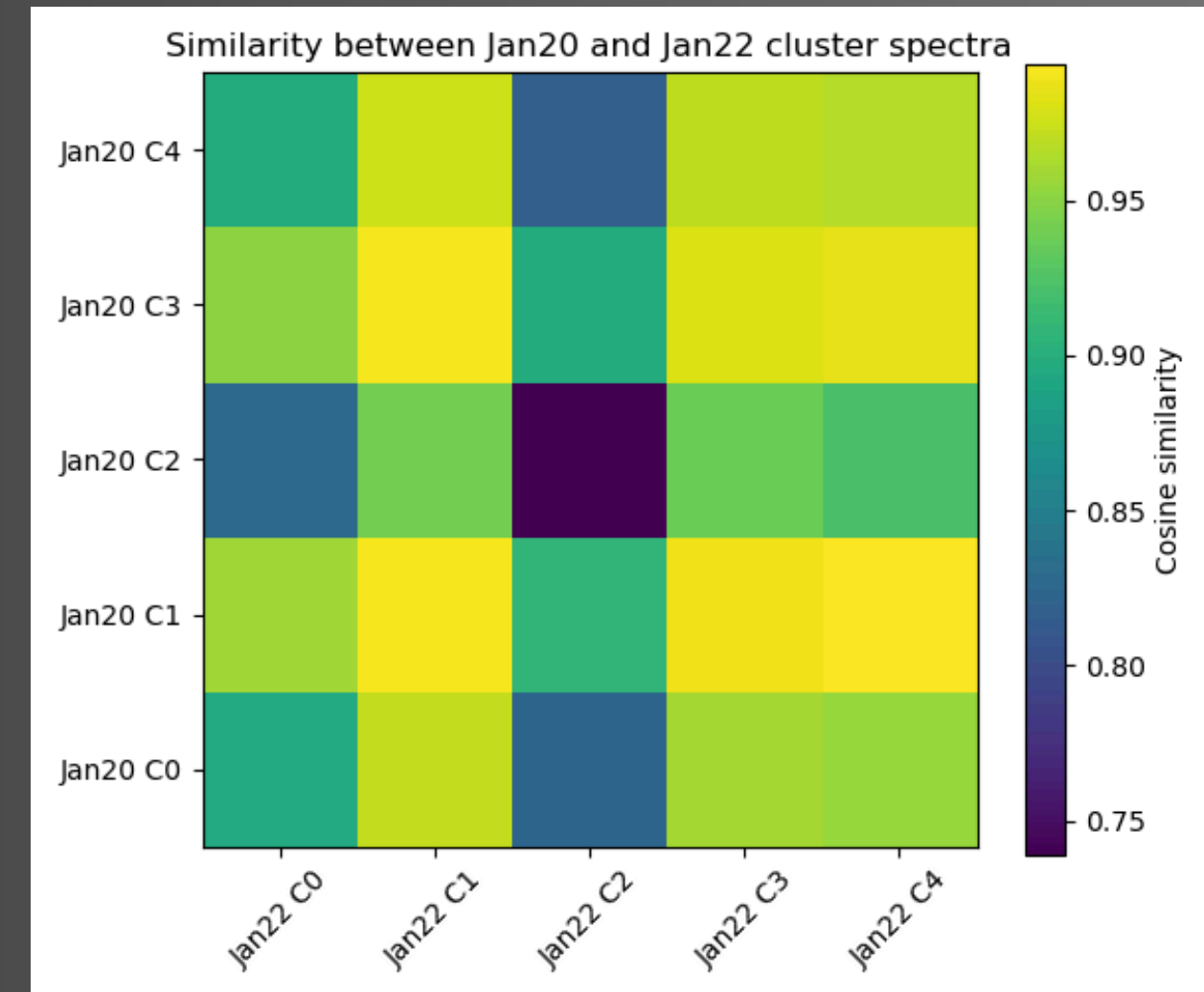
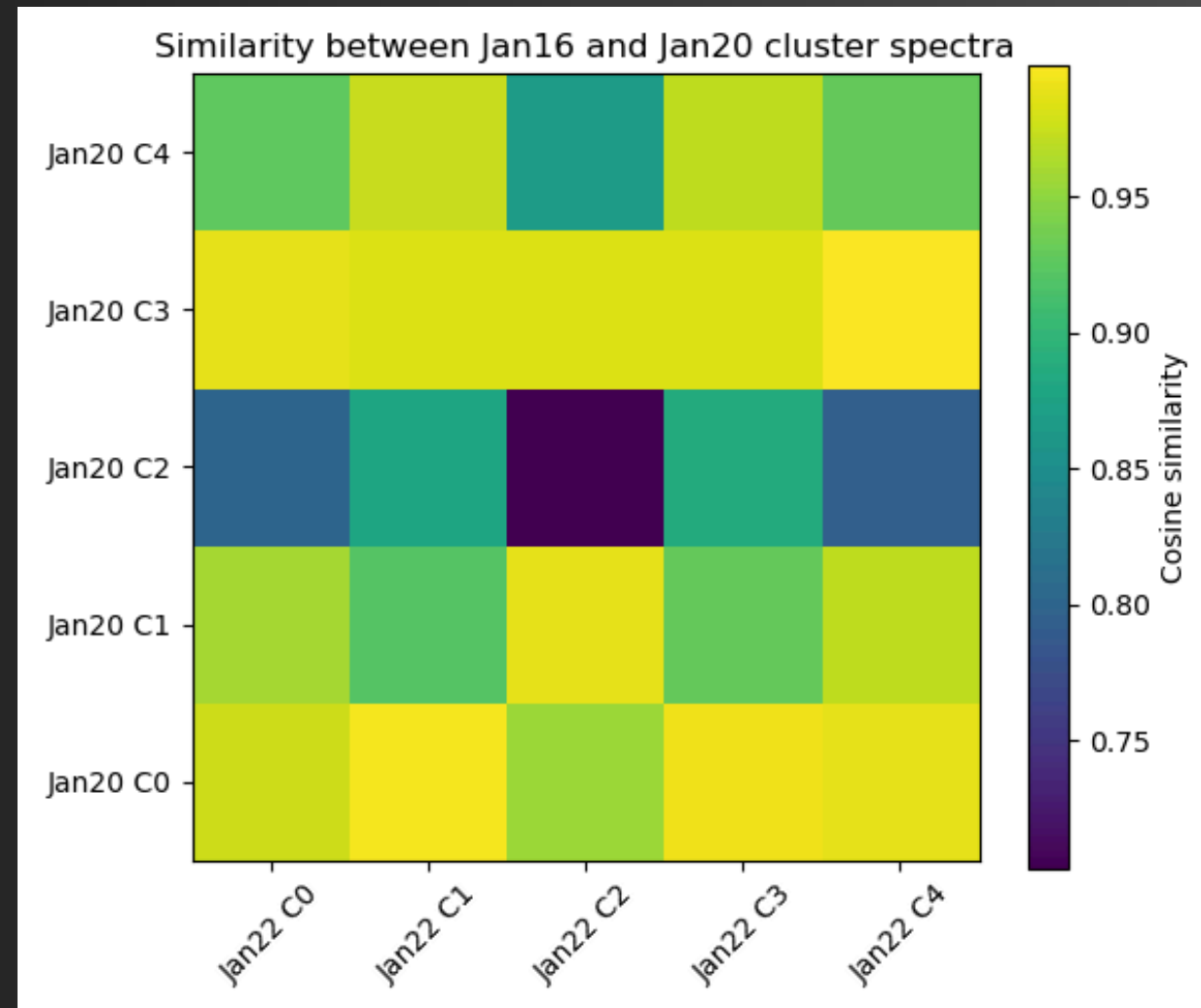
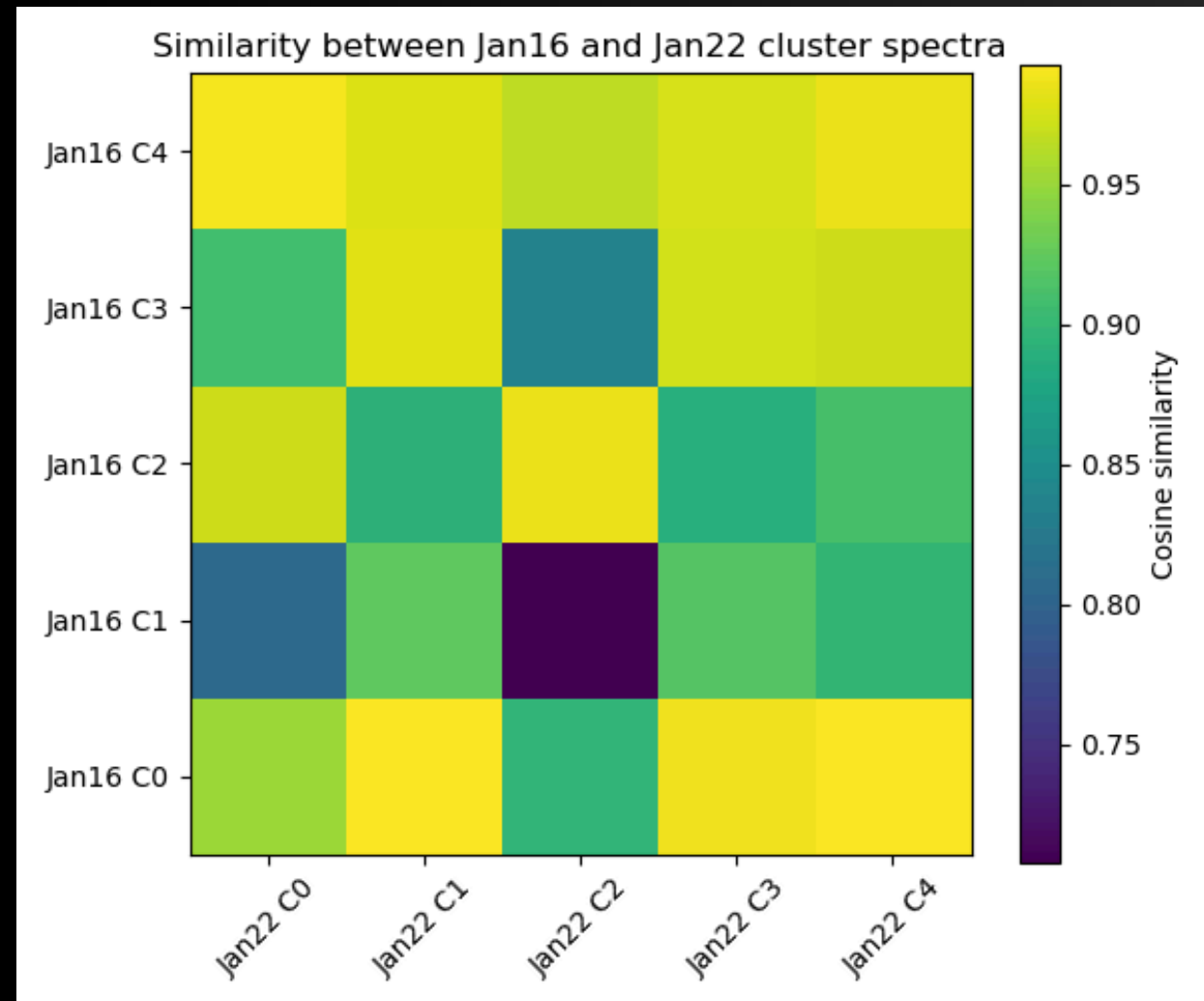
Each station activates different regions of the SOM grid, demonstrating that stations have unique spectral fingerprints, likely reflecting differences in site conditions, local geology, and distance to the epicenter.

Station 0, 1, 2 have a stable, repeatable frequency-domain response!



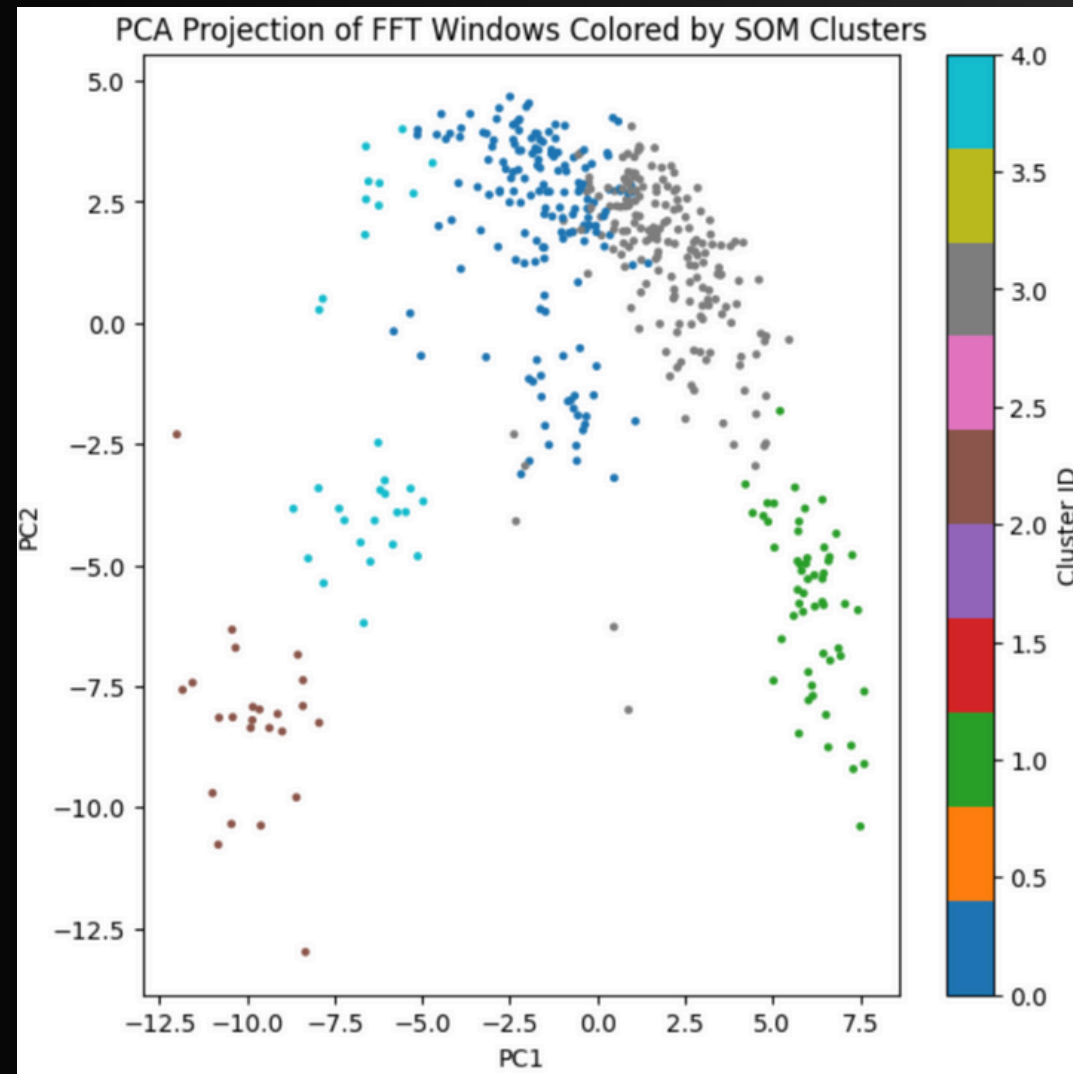
- Stations 0, 1, and 2 show nearly identical hit-map patterns: dominated by local site effects rather than event differences.
- Stations 3 and 4 show more variation: frequency content is more sensitive to event-specific factors.

Similarity Comparisons

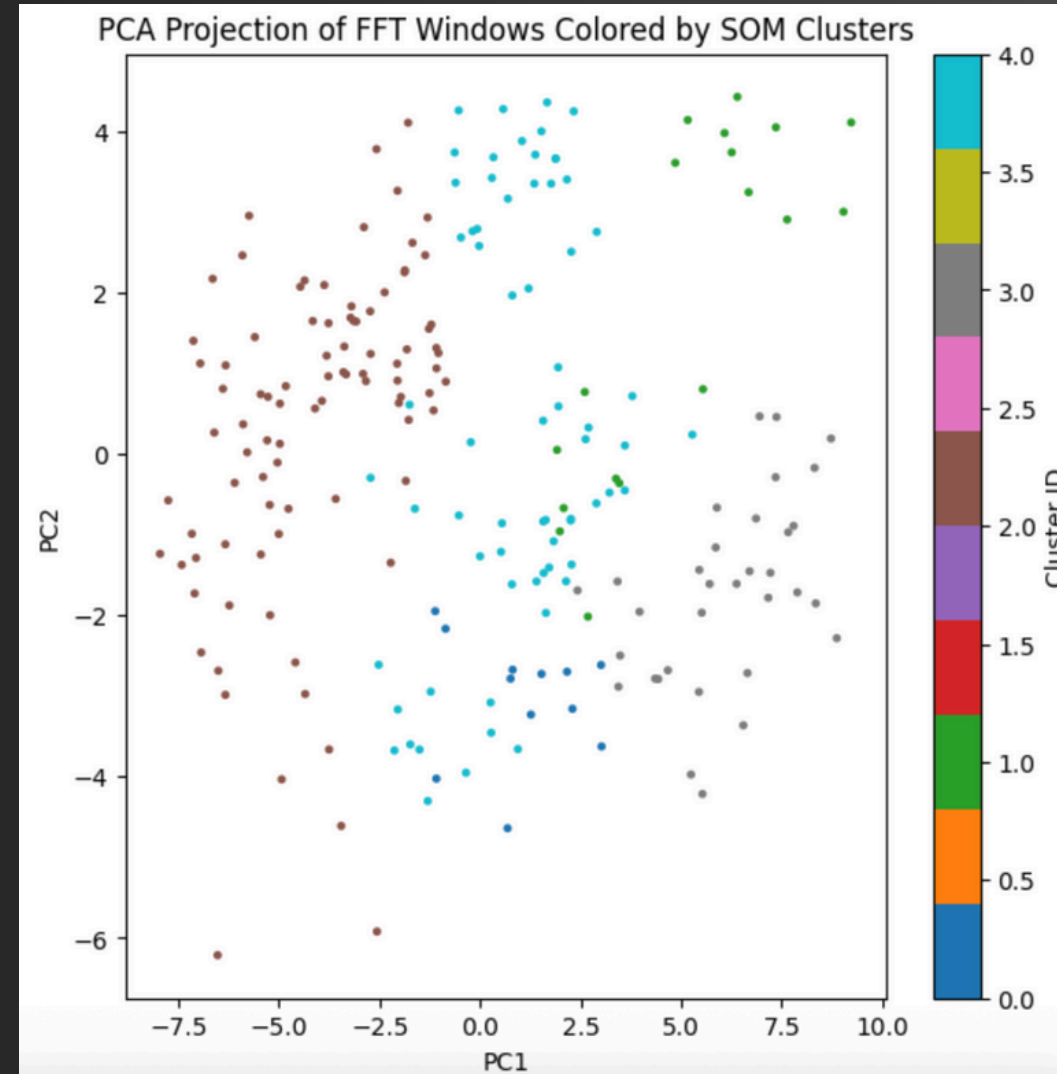


Cosine similarity between cluster spectra shows >0.9 similarity for most cluster pairs, confirming that earthquakes share consistent frequency-domain structure. This indicates that, despite occurring on different days, the overall frequency-domain patterns of the earthquakes remain highly similar. Cluster C0, C1, C3 and C4, represent stable, repeatable spectral patterns. Cluster C2 is the least stable across events, and could likely be our “difference detector” between events.

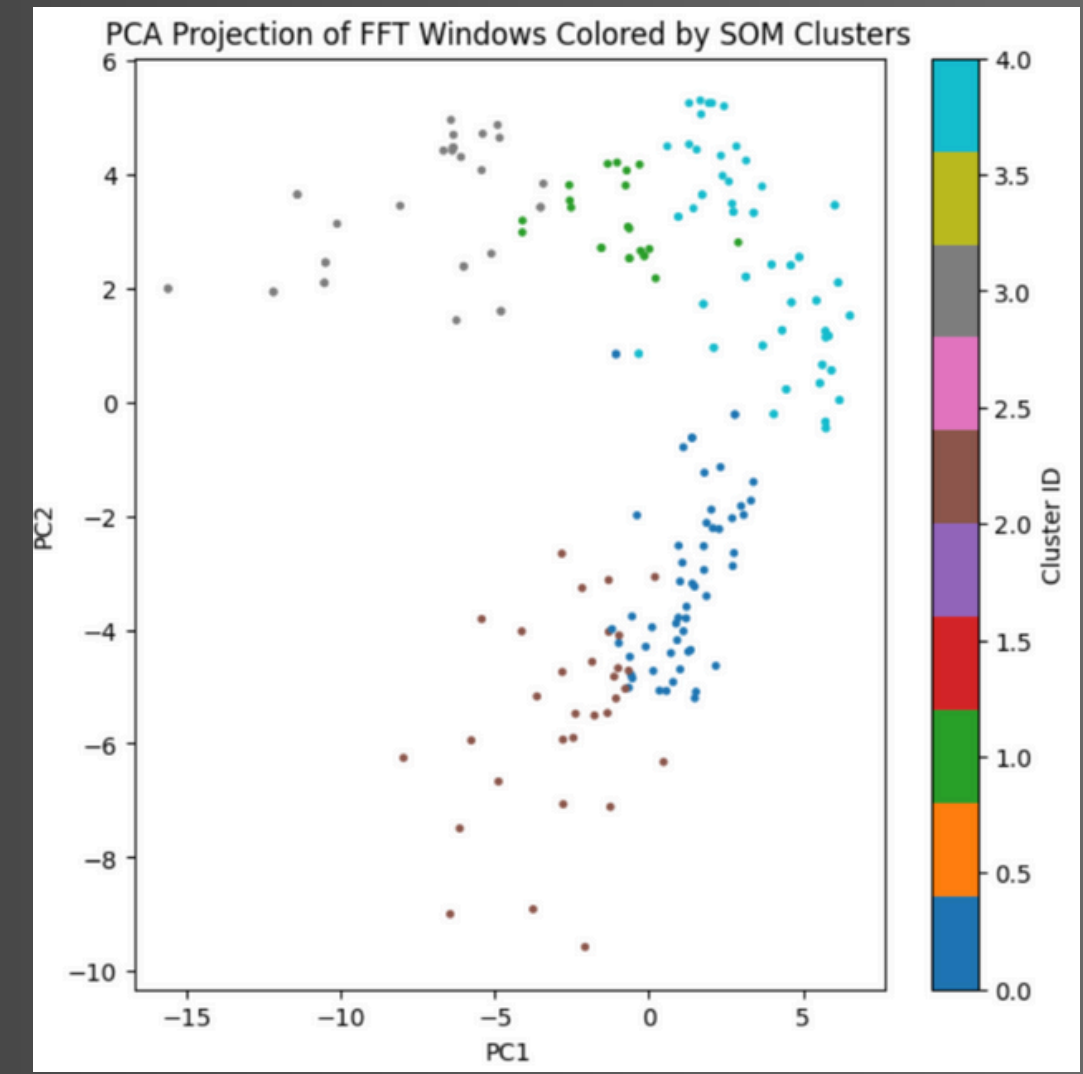
Resulting PC Space Projection



January 16, 2022



January 20, 2022



January 22, 2022

Each event produces clear, coherent spectral clusters, and these clusters consistently occupy similar regions of PC space. This demonstrates that the underlying frequency-domain structure of the earthquakes is stable and repeatable, while the few shifts observed between events reflect natural variations in source characteristics and station responses. Overall, PCA confirms that the SOM clusters capture real, physical similarities in the earthquake spectra across different days.

What does this mean?

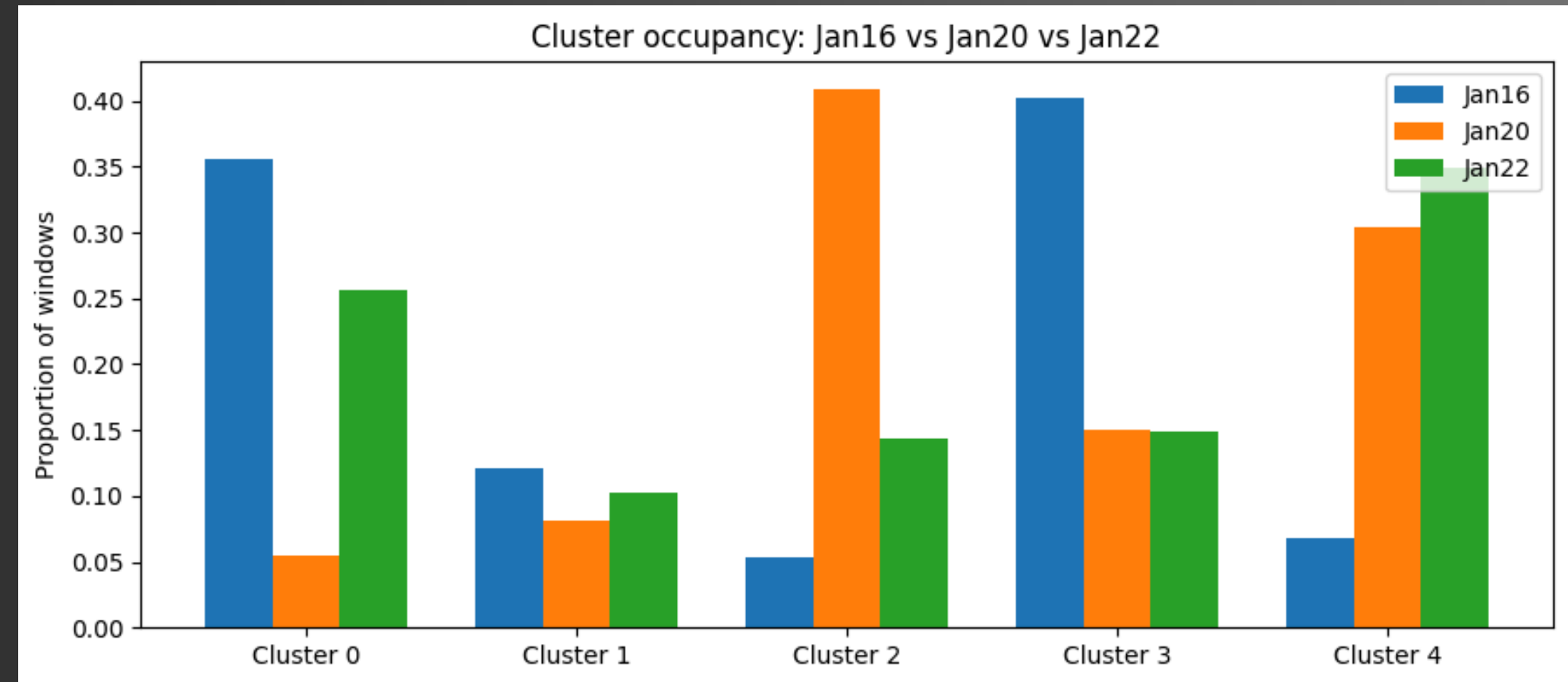
**Stable station behaviour
across earthquakes**

Stations share similar frequency-
domain structure.

**Spectral Clusters are similar
but not identical across events**

All three earthquakes show:

- the same types of spectral patterns (same cluster shapes),
- similar PCA layouts,
- and high similarity scores (0.85–0.97) between clusters.



Each earthquake activates the same clusters, but in different proportions — showing similar spectral patterns with event-specific shifts in how often they occur.

Conclusions

How the results answer the research questions?

- Stations 0, 1, and 2 show stable, repeatable responses across all three earthquakes → stations behave consistently.
- All three earthquakes contain similar frequency-domain patterns, confirmed by SOM clusters, PCA layouts, and high similarity scores.
- Earthquakes differ mainly in how often each cluster appears, revealing event-specific spectral variation.

What we learned about the method?

- FFT + SOM is a powerful combination for analyzing high-dimensional spectral data.
- FFT extracts meaningful spectral features, and SOM organizes those features into clear, interpretable clusters — making it easy to compare stations and earthquakes.
- SOM visualizations (hit maps, U-matrices, PCA colouring) clearly separate station stability from event-level differences.
- The method provides an interpretable way to compare earthquakes beyond simple time-series inspection.

Limitations

- Only three earthquakes were analyzed, so some similarities may be coincidental rather than truly representative.
- Small dataset limits the ability to generalize results to a wider range of events.
- SOM hyperparameters and map size may influence cluster boundaries.

References

- RAI UNAM. (n.d.). Base de Datos de registros acelerograficos de la raii-UNAM. Base de datos de registros acelerograficos de la Red Sismica Mexicana.
<https://aplicaciones.iingen.unam.mx/AcelerogramasRSM/Inicio.aspx>